# Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

## STATEMENT

Data Engineer with 3 years of industry experience building production **Python** and **PostgreSQL** data pipelines in regulated financial services, plus an MSc in Data Science (University of Bristol). Designed a **Normalized SQL Schema** fed by scheduled **API** ingestion that merged in legacy data, and built an **AWS** pipeline (**S3**, **Lambda**, **EventBridge**, **SageMaker**) that parses **300+** daily emails into structured records, saving **60** analyst hours a month. Delivered **Power BI** dashboards to analysts, worked with the security team on access control and network restrictions, and acted as de facto team lead for up to 4 interns. Co-built a preprocessing pipeline with integrity checks and an audit manifest for an MSc dissertation with AstraZeneca.

## EDUCATION

**University of Bristol – MSc. Data Science (awaiting result)** September 2025 – September 2026
**NMIMS B.Tech Information Technology (Dual Degree)** June 2017 – April 2022
**NMIMS MBA Technology Management (Dual Degree)** June 2017 – April 2022

## WORK EXPERIENCE

**HDFC Asset Management Company Ltd.** July 2022 – July 2025

**AI Solutions Engineer / Forward Deployed Engineer**

### Data Engineering & Pipelines

- Designed a **Normalized SQL Schema** (**PostgreSQL**) to the analysts' requirements and wrote **Python** ingestion scripts for the **Morningstar API**, covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly (fixed income) and monthly (equity); integrated the legacy analyst data previously held in Excel into the same database and developed **Optimized SQL** queries, **Functions** and **Procedures** for efficient data insertion.
- Built a **Python** pipeline that parses over **300** daily emails from credit rating agencies into structured data, **fuzzy-matches** company names against the ones analysts track and sends a daily consolidated email with an Excel copy stored in **S3**, **saving 60 analyst hours a month**; first built on **AWS Textract**, then moved to **Azure OpenAI** when an agency began sending emails as images and broke the parser; scheduled at 9 AM through **EventBridge**, **Lambda** and **AWS SageMaker**.
- Analyzed customer data with **Python Pandas** and **Market Basket Analysis** (**Apriori**) for the marketing team; migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**, and recreated them in the **ReactJS** frontend so the wider organization could use the data without Power BI licences.

### Cloud, Security & Systems Integration

- Designed the **AWS** infrastructure for the analytics platform and applications (**EC2**, **S3**, **ECR**, **Lambda**), provisioned **EC2** instances using **CloudFormation** templates prebuilt by the security team, and packaged the **FastAPI** microservices as **Docker** containers behind an **Nginx** reverse proxy.
- Automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.
- Worked under security and compliance constraints: **RBAC** with a **Python** authentication function against the company **LDAP** server, **AWS Security Groups** limiting access to company private IPs, and coordination with the security team on port whitelisting; the **Nginx** proxy was one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.

### GenAI Applications & Stakeholders

- Designed and built **LLM applications** end to end (**ReactJS** frontend, **Python FastAPI** microservices, **RAG** with **LlamaIndex** and **Azure OpenAI**) with a separate **pgvector** database per department; **tool-calling** agents run **SQL** queries over the fund data to compare funds and generate PDF reports on request.
- Worked directly with Client Service, Compliance and Investment from requirements to release; built the first systems as the only engineer alongside my manager, later acted as de facto team lead for up to 4 interns; recognised with Best Team Project twice (2024 rating-email automation, 2025 department RAG chatbot).

## PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; personal focus on the data preprocessing, scoring algorithm and RAG assistant. Preprocessing pipeline that joins 17 raw and augmented inputs (DepMap, GEO, HPA) into 14 clean tables (CSV and Parquet), with identity resolution across sources, duplicate and missing-value handling, integrity checks and a build manifest audit file; scoring ranks cancer cell lines by target inclusion and exclusion genes across six evidence layers; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Study comparing CNN/ViT/ResNet encoders with symbolic, SBERT embedding-retrieval and LSTM sequence decoders for image descriptions on three synthetic datasets, over five seeds, scored with SBERT cosine similarity, BLEU and ROUGE-L and compared with BLIP/SigLIP and commercial VLMs, noting where the evaluation protocols differed.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Wingman:** AI dating-coach web app built end to end (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM) with 108 LLM personas across 12 archetypes and a vision model scoring uploaded photos. Rate limits shaped the design: 12 calls instead of 108, serialised calls, results streamed to the UI and written to disk incrementally; LLM output treated as untrusted (validated with Pydantic, repaired) and cached by input hash. GitHub: https://github.com/Arishtanemi4/wingman.git
- **Swindon Borough Council Fuel-Poverty Hack:** A household and neighborhood-based fuel poverty data analysis tool. | GitHub: https://github.com/Arishtanemi4/fuel-poverty.git

## ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

## SKILLS

- **Tools:** AWS (EC2, S3, ECR, Lambda, EventBridge, SageMaker, CloudFormation templates), Docker, Jenkins, GitLab (CI/CD), Power BI, Azure OpenAI, LlamaIndex, Nginx, Linux, LDAP, Git, VS Code, Claude Code.
- **Languages:** Python (pandas, FastAPI, Pydantic, scikit-learn, fuzzywuzzy), SQL, R (dplyr, tidyverse, ggplot2), JavaScript/TypeScript (NodeJS express, ReactJS), Shell
- **Databases:** PostgreSQL (schema design, SQL functions and procedures, pgvector), SQLite